Assignment 6

Q1 (a)

$$
\begin{aligned}
-1(1+e^{-x})^{-2} &= \frac{e^{-x}}{1+e^{-x}}\frac{1}{1+e^{-x}} \\
&= \frac{e^{-x}}{1+e^{-x}}\frac{1}{1+e^{-x}}
\end{aligned} \tag{1}
$$

but notice

$$\frac{e^{-x}}{1+e^{-x}} = 1 - \frac{1}{1+e^{-x}} \tag{2}$$

$$\frac{d}{dx}\sigma(x) = -1(1+e^{-x})^{-2} = 1 - \frac{1}{1+e^{-x}} \tag{3}$$

Q2